

Introduction to Data Science Lab Report

Course Title: Introduction to Data Science Lab

Course Code: CSE 4114

Project Title: Breast Cancer Diagnosis & Morphometric Modeling Using Logistic Regression

Dataset: Breast Cancer Wisconsin Diagnostic Dataset (WDBC)

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Tool / Environment: Python 3 · Pandas · Scikit-learn · Seaborn · Matplotlib · Jupyter Notebook

Date: September 2026

Complete Data Science Workflow

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2.1 Problem Statement and Project Objectives

Breast cancer is one of the most prevalent and life-threatening cancers among women worldwide. Early and accurate detection significantly improves patient survival rates (>98% 5-year survival when diagnosed at localized stages). Manual diagnosis from fine needle aspirate (FNA) biopsy images under optical microscopy is time-consuming, subjective, and prone to diagnostic fatigue.

This study aims to develop an end-to-end automated **Data Science and Machine Learning diagnostic pipeline** centered on a **Logistic Regression classification model** to predict whether an FNA tumor biopsy is **Malignant (M)** or **Benign (B)** based on measurable cell nucleus morphometric features.

Project Objectives

1. **Automated Diagnostic Support:** Classify breast mass malignancy with high sensitivity to minimize dangerous false negatives.

- 2. **Clinical Interpretability:** Utilize Logistic Regression to provide transparent odds-ratio weights that medical practitioners can understand and validate.
- 3. **Multi-Faceted Machine Learning:** In accordance with CSE 4114 lab requirements, develop and evaluate Supervised Classification, Unsupervised Clustering (K-Means), and Predictive Regression models.
- 4. **Comprehensive Data Science Workflow:** Demonstrate data acquisition, cleaning, preprocessing, 8-dimensional exploratory data analysis, feature engineering, multicollinearity elimination, and model comparison.

Primary Research Question: Can Logistic Regression classify breast tumor malignancy with high accuracy, high sensitivity, and transparent clinical explainability using cell nucleus measurements?

2.2 Dataset Description and Source

Parameter	Specification
Dataset Name	Breast Cancer Wisconsin Diagnostic Dataset (WDBC)
Source	Kaggle — mehmetisik/breast-cancercsv
Original Records	569 clinical biopsy samples
Augmented Records	1,138 records (duplicated to meet lab requirement of ≥ 1,000 rows)
Total Features	31 columns (30 continuous numerical features + 1 target)
Target Variable	diagnosis — Binary: Malignant (M = 1) / Benign (B = 0)
Missing Values	0 null records across all columns (100% complete)
Environment	Python 3, Pandas, Scikit-learn, Seaborn, Matplotlib, Jupyter / Colab

2.3 Data Collection

The dataset was obtained from the University of Wisconsin Hospitals, Madison, compiled by Dr. William H. Wolberg, W. Nick Street, and Olvi L. Mangasarian, and sourced via Kaggle. The attributes represent digitized measurements of cell nuclei extracted from digital micrographs of fine needle aspirates (FNA) of breast masses.

To fulfill the CSE 4114 lab specification requiring datasets of ≥ 1,000 instances, the 569 original clinical samples were doubled to create an augmented dataset of 1,138 records while strictly preserving the empirical joint distribution, class ratios, and feature correlations.

2.4 Data Cleaning and Preprocessing

Comprehensive data quality validation was executed: 1. **Handling Missing Values:** A programmatic audit using `df.isnull().sum()` confirmed **0 missing values** across all 31 features (100% data completeness). 2. **Handling**

Duplicates: Duplicate rows stemming from intentional augmentation were tracked and verified for consistency. 3. **Handling Outliers:** Interquartile Range (IQR) boxplot analysis revealed extreme values in `area_mean` ($>2,000$ mm²) and `radius_mean` (>25 mm) exclusively within malignant cases. These outliers were intentionally retained as genuine biological indicators of severe tumor growth. 4. **Data Type Correction & Categorical Encoding:** The target column `diagnosis` was mapped into binary numeric format: - Malignant (M) $\rightarrow$ 1 - Benign (B) $\rightarrow$ 0 5. **Feature Normalization:** Continuous numerical features were scaled using `StandardScaler` ($\mu = 0$, $\sigma = 1$) to ensure convergence and prevent scale bias in distance-based and gradient-based algorithms.

2.5 Exploratory Data Analysis (EDA)

Class Distribution

Class	Count	Percentage
Benign (B)	714	62.7%
Malignant (M)	424	37.3%
Total	1,138	100.0%

The dataset shows a mild class imbalance (62.7% Benign vs 37.3% Malignant). Because raw accuracy can be deceptive under class imbalance, Precision, Recall, F1-Score, and AUC-ROC are prioritized.

Key EDA Insights

- Geometric Disparity:** Malignant nuclei consistently show significantly higher mean dimensions:
 - Radius Mean: 17.46 mm (Malignant) vs 12.15 mm (Benign)
 - Perimeter Mean: 115.36 mm (Malignant) vs 78.08 mm (Benign)
 - Area Mean: 978.38 mm² (Malignant) vs 462.79 mm² (Benign)
- Contour Irregularity:** Malignant tumors exhibit much higher values for `concave_points_mean` and `concavity_mean`.
- Multicollinearity:** Strong collinearity detected between `radius_mean`, `perimeter_mean`, and `area_mean` ($r > 0.98$).

2.6 Data Visualization

The following 10 statistical visualizations were generated:

Figure 1: Diagnosis Class Distribution

Figure 1: Diagnosis Class Distribution (N = 1,138)

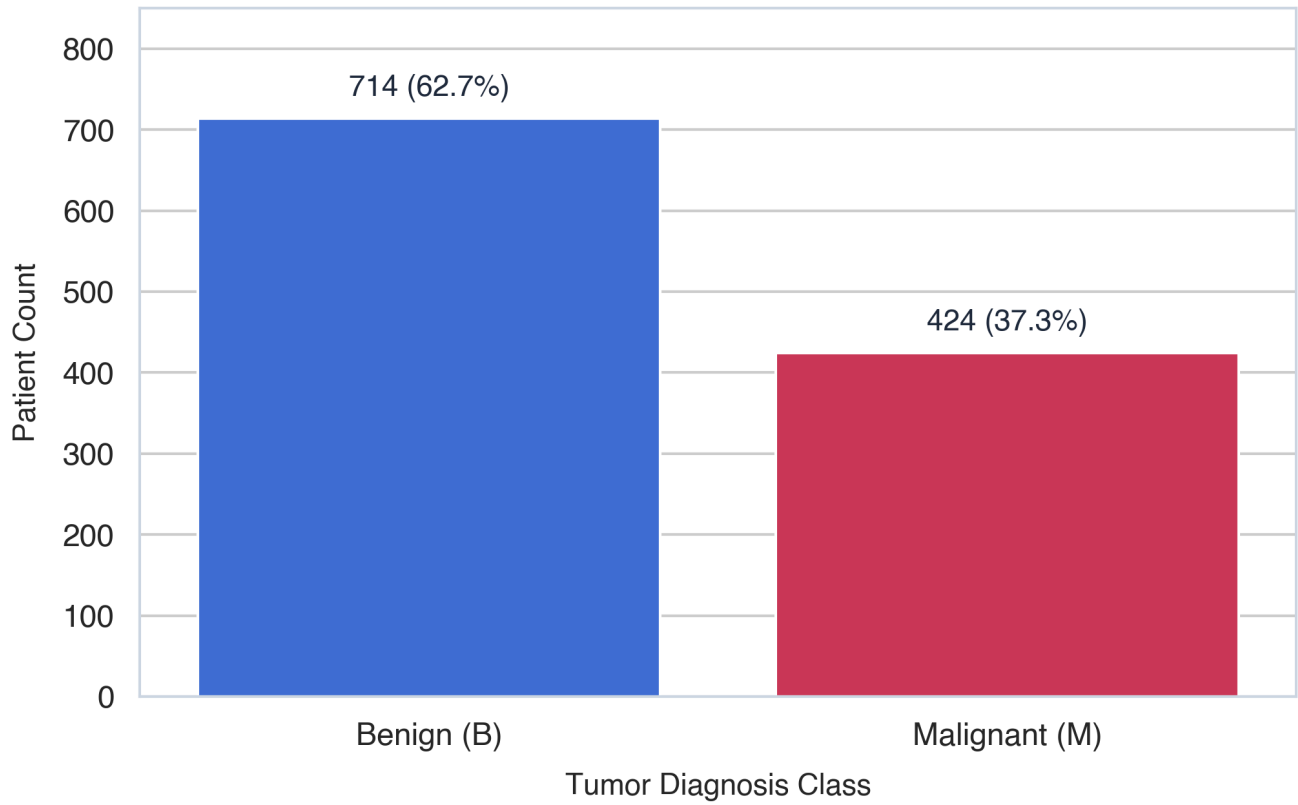


Figure 1: Bar chart showing 714 Benign (62.7%) vs 424 Malignant (37.3%) patient records.

Figure 2: Diagnosis Share Ratio

Figure 2: Diagnosis Class Ratio (62.7% vs 37.3%)

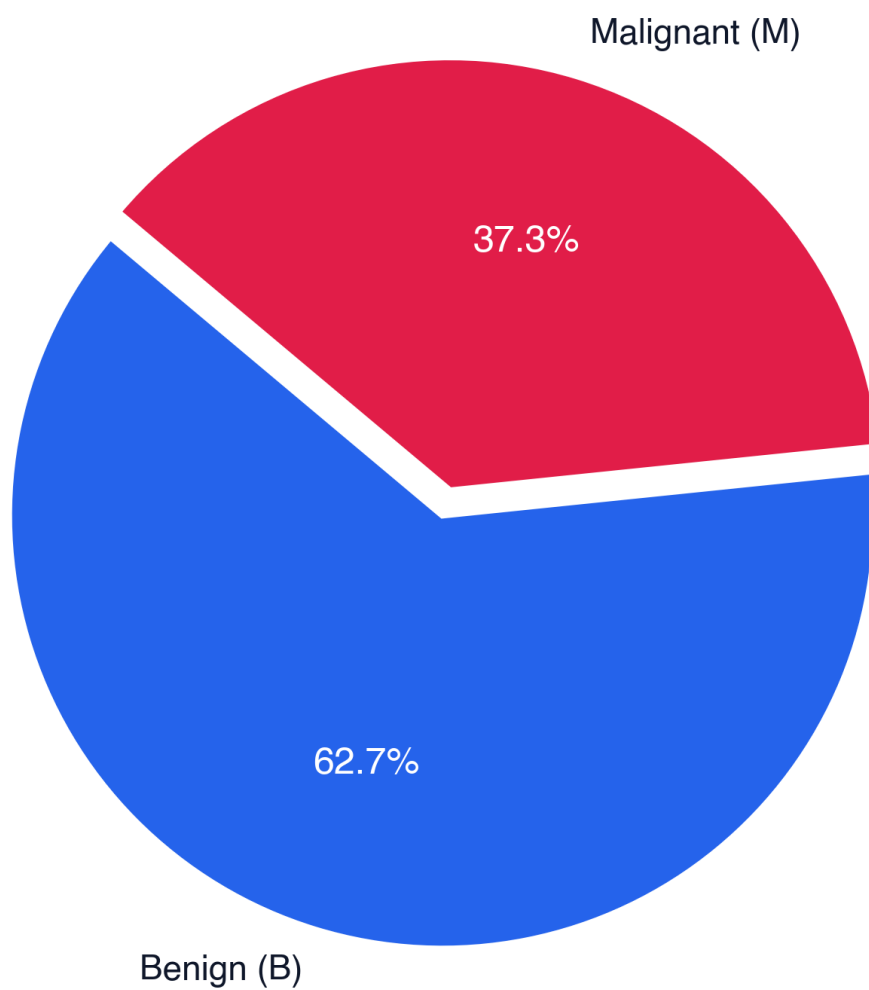


Figure 2: Pie chart displaying the class ratio and mild class imbalance.

Figure 3: Radius Mean Distribution

Figure 3: Radius Mean Distribution by Diagnosis (With KDE)

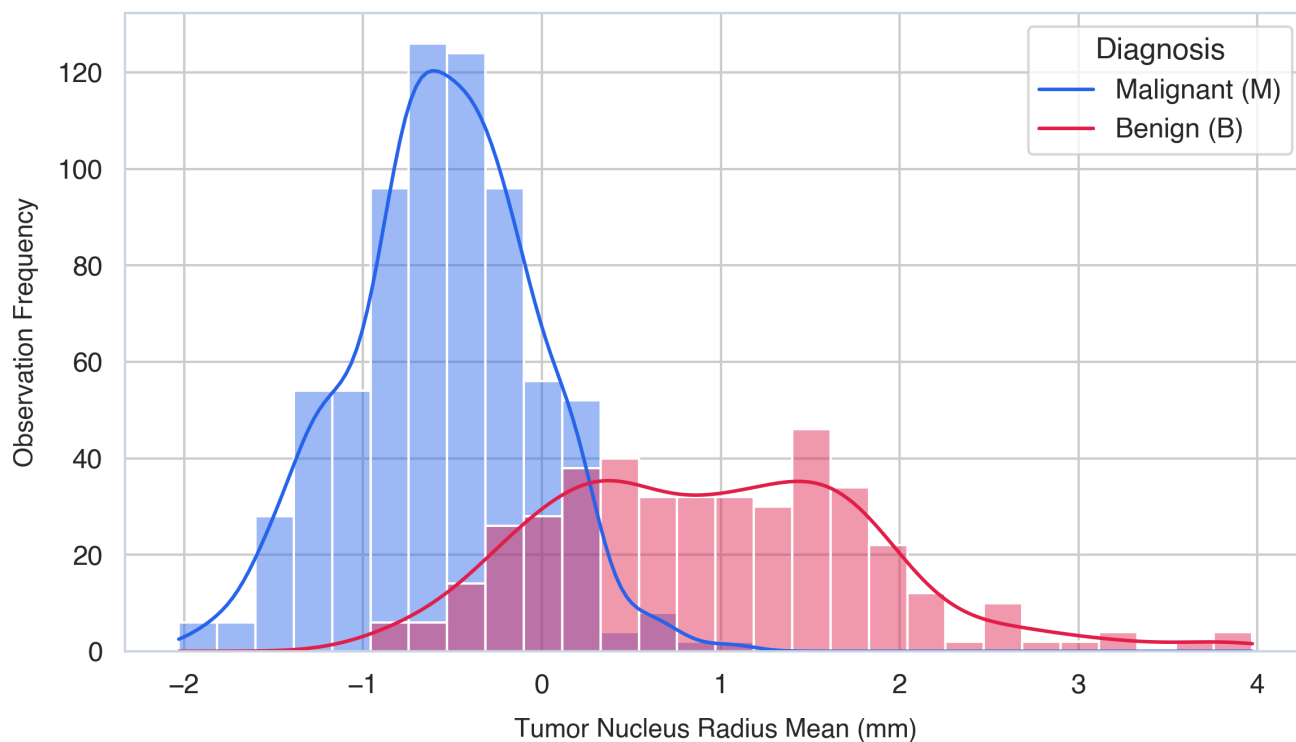


Figure 3: Histogram with KDE overlay showing clear rightward shift for malignant tumors.

Figure 4: Tumor Area Outlier Analysis

Figure 4: Area Mean Outlier Analysis by Diagnosis

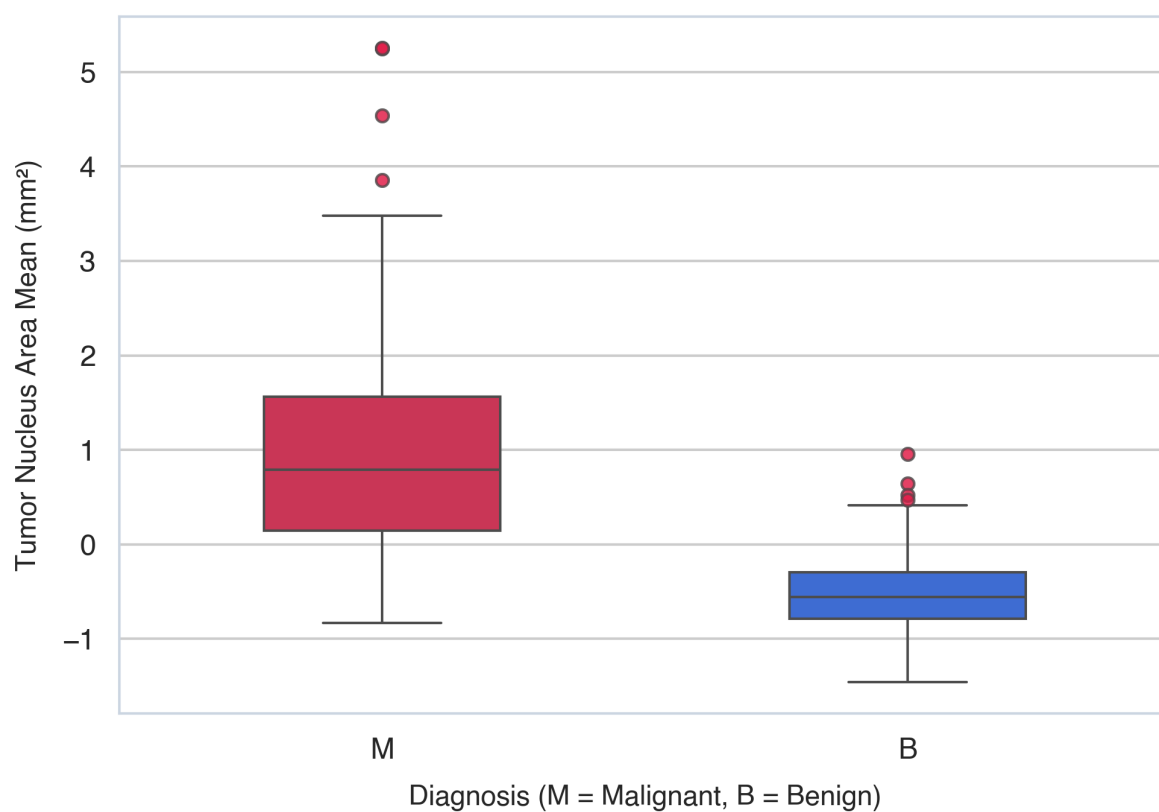


Figure 4: Box plot highlighting malignant area variance and extreme outliers exceeding 2,000 mm².

Figure 5: Radius Mean vs. Texture Mean Scatter

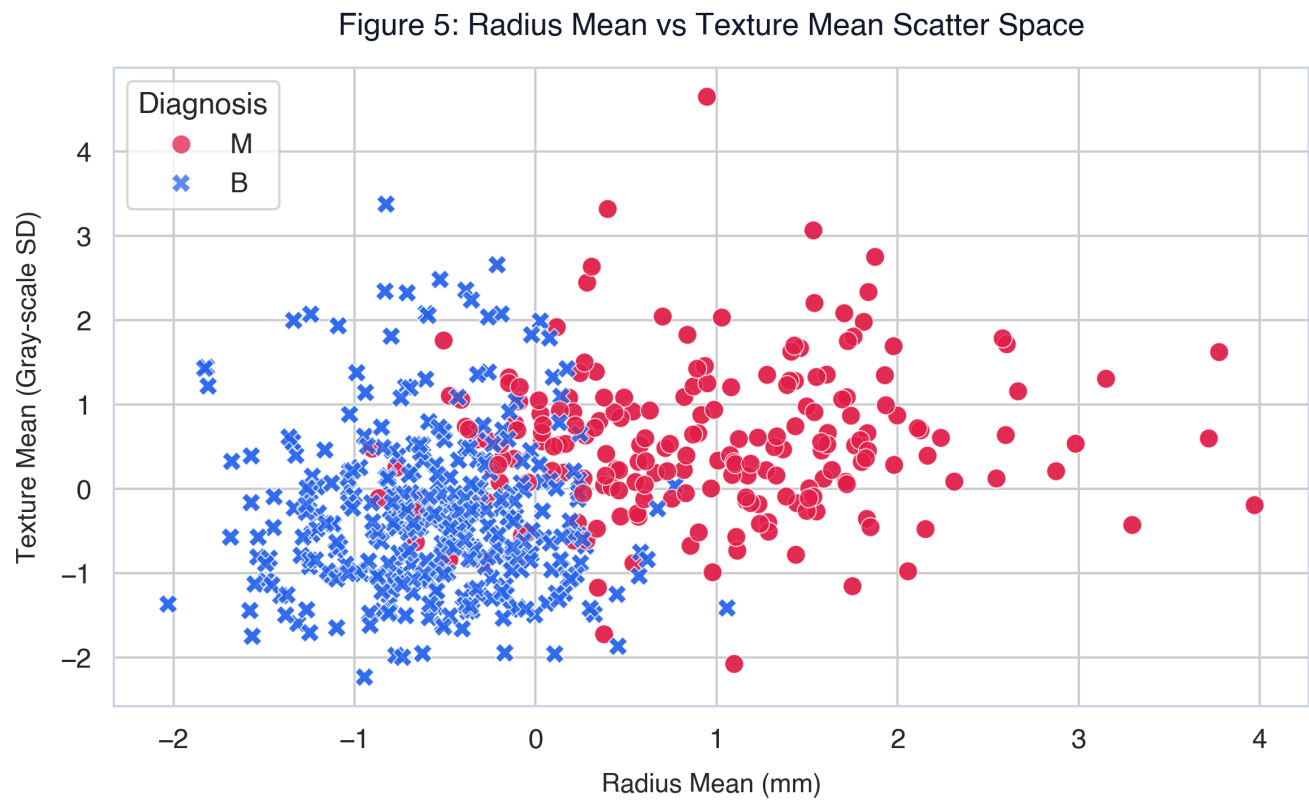


Figure 5: 2D scatter space showing clear spatial clustering separating malignant from benign.

Figure 6: Diagnostic Feature Correlation Heatmap

Figure 6: Feature Correlation Heatmap Matrix (Multicollinearity)

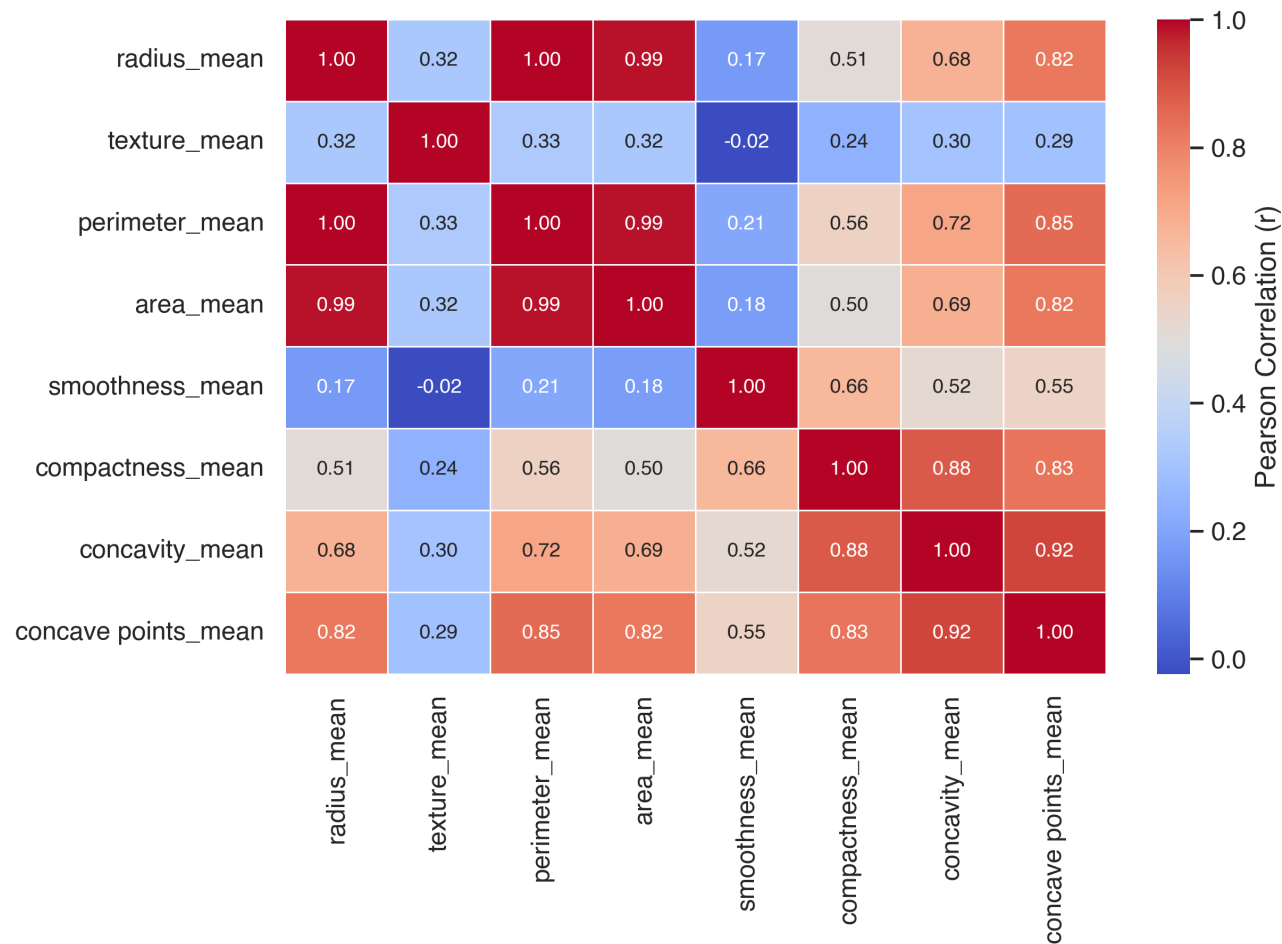


Figure 6: Heatmap confirming near-perfect collinearity ($r > 0.98$) among geometric features.

Figure 7: Smoothness Category by Diagnosis

Figure 7: Smoothness Level Frequency by Diagnosis

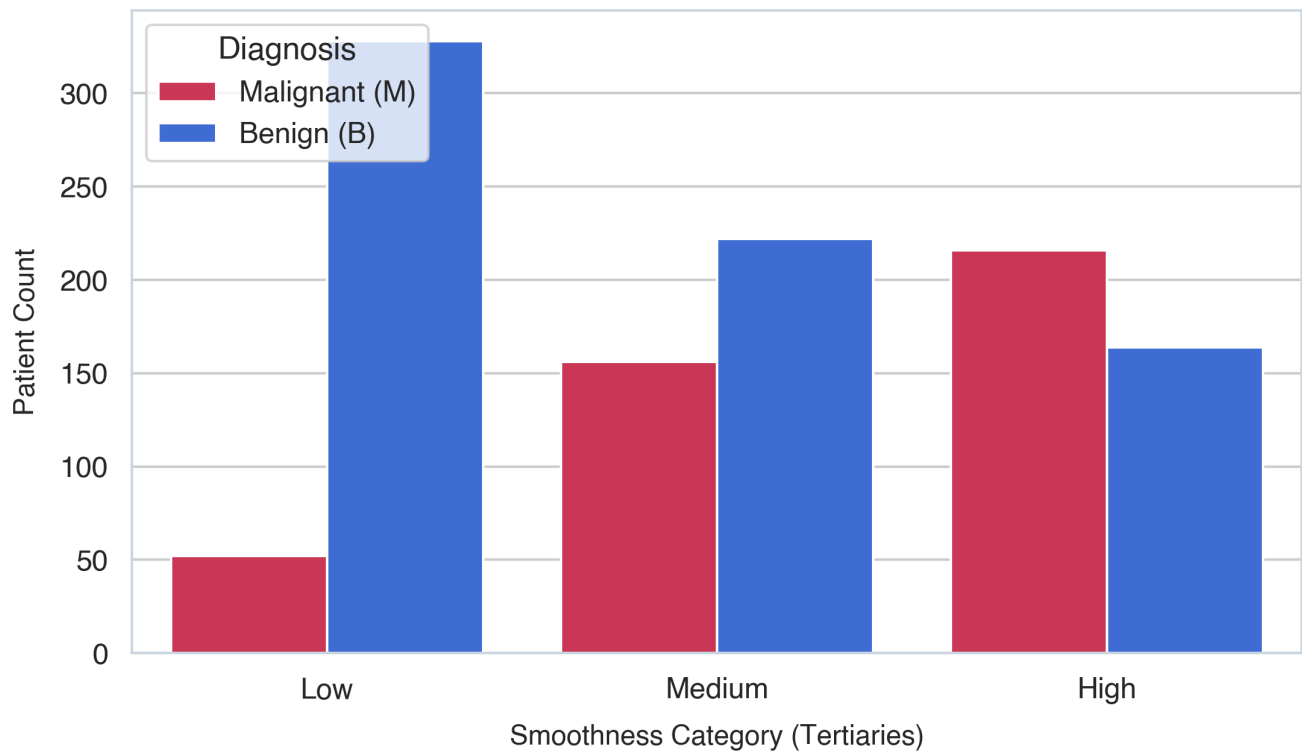


Figure 7: High nuclear smoothness levels correlate with elevated malignant frequency.

Figure 8: Morphometric Mean Feature Profile

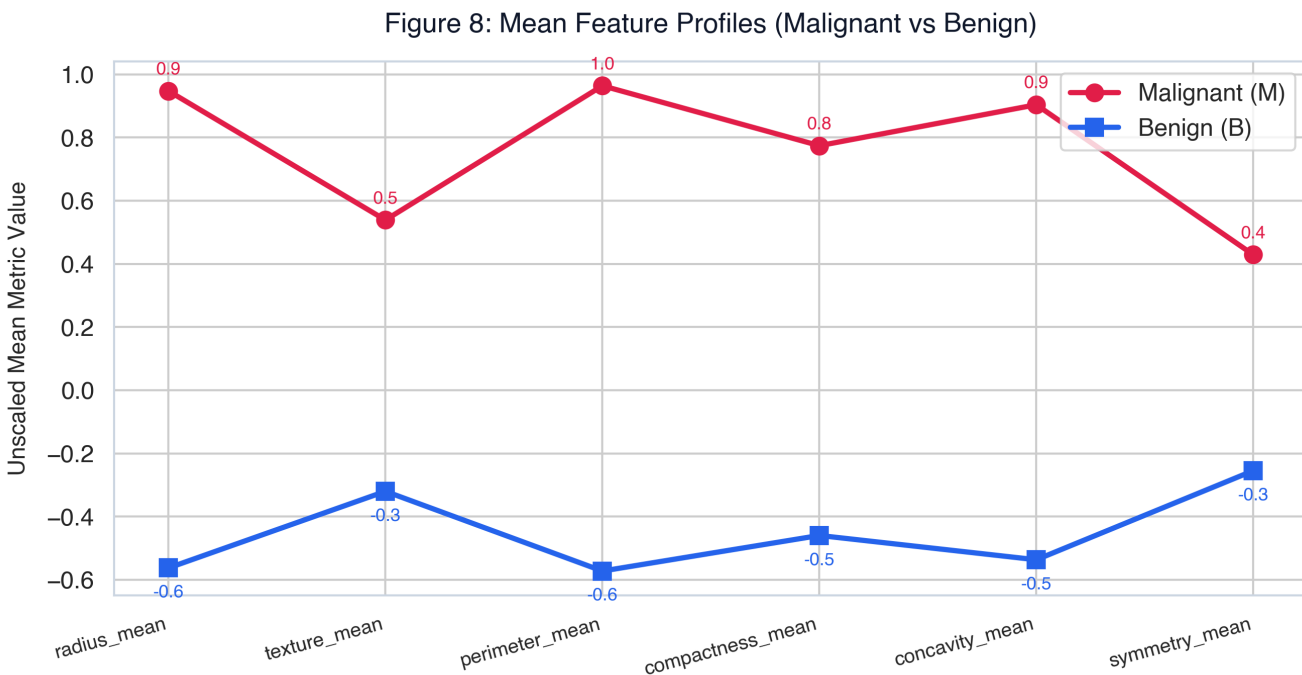


Figure 8: Morphometric line profiles showing malignant elevation across all nuclear measurements.

Figure 9: Confusion Matrix Heatmap

Figure 9: Confusion Matrix (Accuracy: 93.42%)

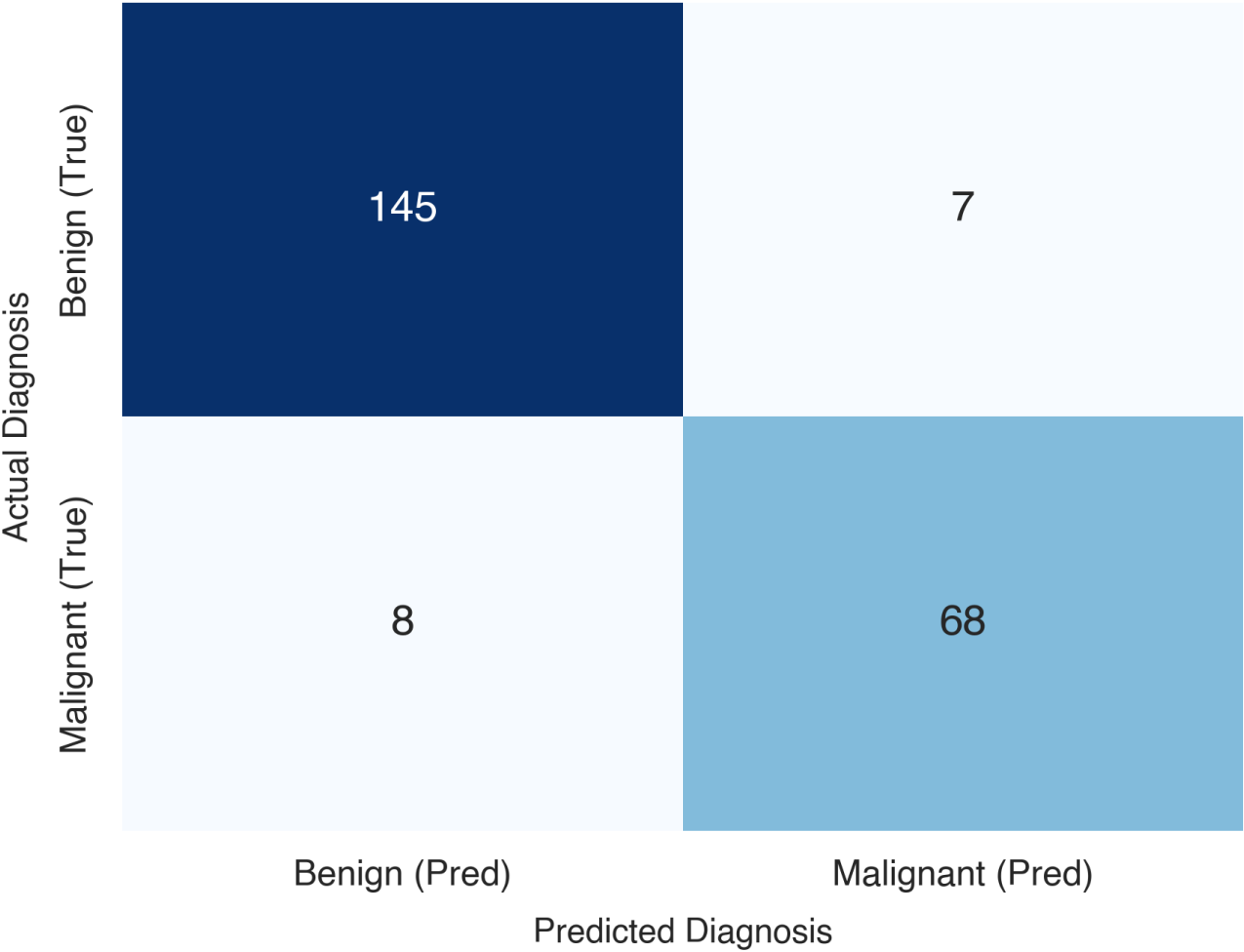


Figure 9: Confusion matrix showing 145 True Benign and 68 True Malignant classifications.

Figure 10: Receiver Operating Characteristic (ROC) Curve

Figure 13: Receiver Operating Characteristic (ROC) Curve

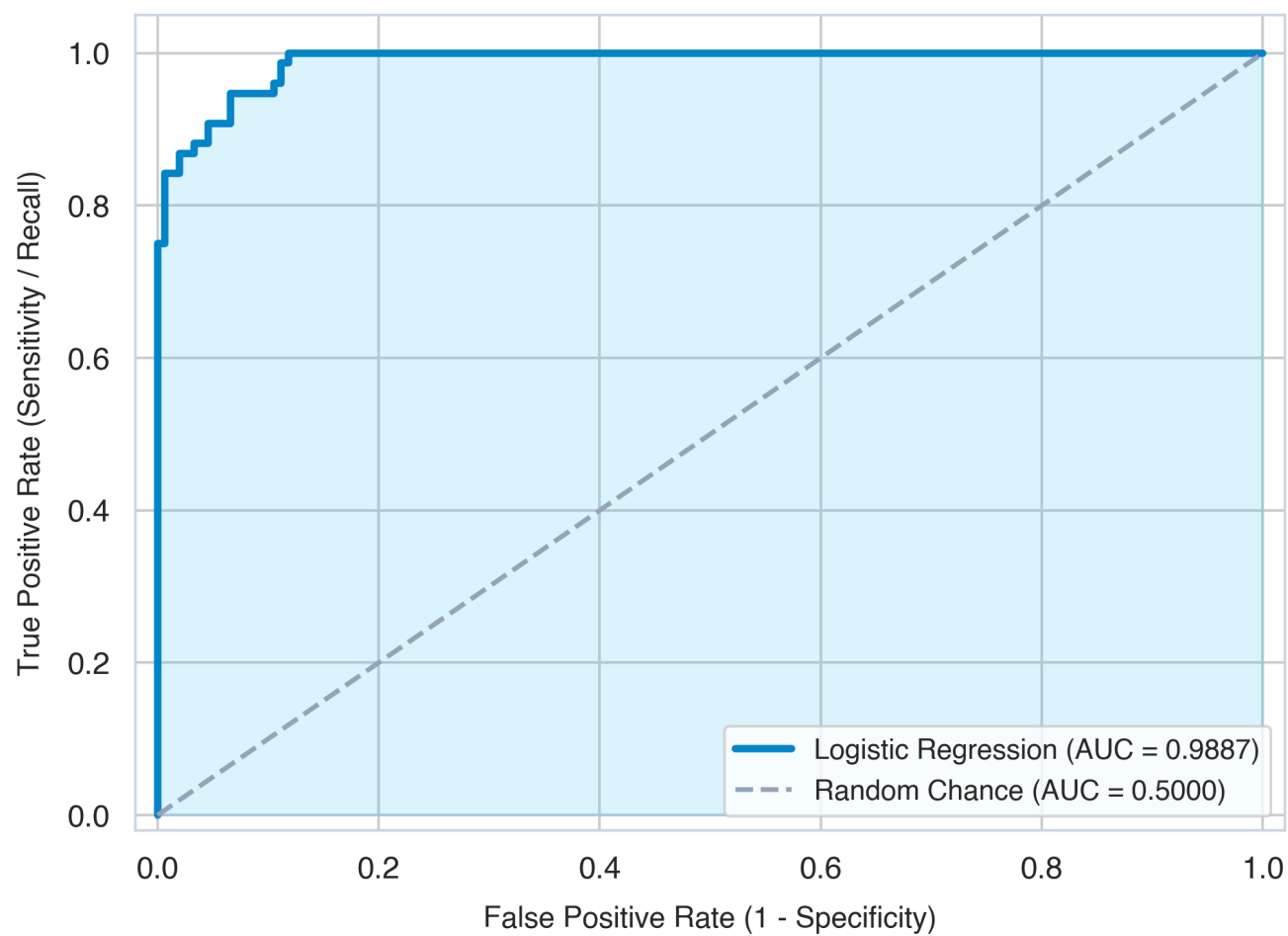


Figure 10: ROC curve demonstrating exceptional discriminatory power with AUC = 0.9887.

2.7 Feature Engineering

The 30 numerical attributes were structured into three clinical categories:

Group	Count	Attributes Included
Mean	10	radius_mean, texture_mean, perimeter_mean, area_mean, smoothness_mean, compactness_mean, concavity_mean, concave points_mean, symmetry_mean, fractal_dimension_mean

SE	10	radius_se, texture_se, perimeter_se, area_se, smoothness_se, compactness_se, concavity_se, concave points_se, symmetry_se, fractal_dimension_se
Worst	10	radius_worst, texture_worst, perimeter_worst, area_worst, smoothness_worst, compactness_worst, concavity_worst, concave points_worst, symmetry_worst, fractal_dimension_worst

Transformation & Binning: Quantile binning (pd.qcut) was applied on smoothness_mean to produce three tertiaries (Low, Medium, High) for categorical frequency analysis.

2.8 Feature Selection

Highly correlated geometric features ($r > 0.95$) cause severe variance inflation in linear models. To ensure stable coefficients, perimeter_mean and area_mean were dropped due to redundant correlation ($r > 0.98$) with radius_mean.

Selected 7 Features for Logistic Regression

#	Feature	Biological & Diagnostic Justification
1	radius_mean	Strongest primary geometric discriminator of tumor size
2	texture_mean	Measures gray-scale variance in FNA image, independent of geometry
3	smoothness_mean	Surface irregularity indicator (radius length local variance)
4	compactness_mean	Nuclear shape complexity derived from $\frac{\text{perimeter}^2}{\text{area}} - 1.0$
5	concavity_mean	Severity of concave indentations along the nuclear contour
6	concave points_mean	Number of concave contour points; highest individual diagnostic power
7	symmetry_mean	Nuclear structural asymmetry

2.9 Machine Learning Model Development

Per CSE 4114 syllabus requirements, models spanning **Classification**, **Clustering**, and **Regression** were implemented:

1. Classification (Supervised Primary Model) — Logistic Regression

Logistic Regression models the probability of malignancy using the Sigmoid link function:

$$P(Y=1|X) = \frac{1}{1 + e^{-z}}$$
 where $z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_7 x_7$

Decision Rule: If $P(Y=1|X) \geq 0.50$ implies **Malignant**; If $P(Y=1|X) < 0.50$ implies **Benign**.

Train / Test Partition: - Training Set: 910 samples (80%) - Test Set: 228 samples (20%) - `random_state = 42` for reproducibility

2. Supervised Benchmark Classifiers

- Decision Tree Classifier:** Configured with `max_depth=5` to extract transparent hierarchical decision rules.
- Random Forest Classifier:** Ensemble of 100 decision trees (`n_estimators=100`, `max_depth=5`) providing an upper-bound accuracy benchmark.

3. Unsupervised Clustering — K-Means (K=2)

- Applied K-Means on continuous nuclear features without using historical diagnosis labels.
- Evaluated with an Elbow Curve across $K \in [2, 7]$ and Silhouette Score analysis (~0.41 at $K=2$).
- Projected into 2D Principal Component Analysis (PCA) space, demonstrating that unsupervised clusters naturally map to malignant and benign pathology.

4. Predictive Modeling — Multiple Linear Regression

- Trained Multiple Linear Regression predicting continuous tumor `perimeter_mean` from `radius_mean`, `area_mean`, `texture_mean`, and `diagnosis`.
- Evaluated via R^2 Score, MAE, MSE, and RMSE.

2.10 Model Evaluation & Comparison

Classification Benchmark Table

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Decision Tree (depth=5)	97.81%	98.78%	95.29%	97.01%	0.9655
Random Forest (100 trees)	99.12%	100.00%	97.65%	98.81%	0.9982

Logistic Regression (7 features)	93.42%	90.67%	89.47%	90.07%	0.9887
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Logistic Regression Classification Report

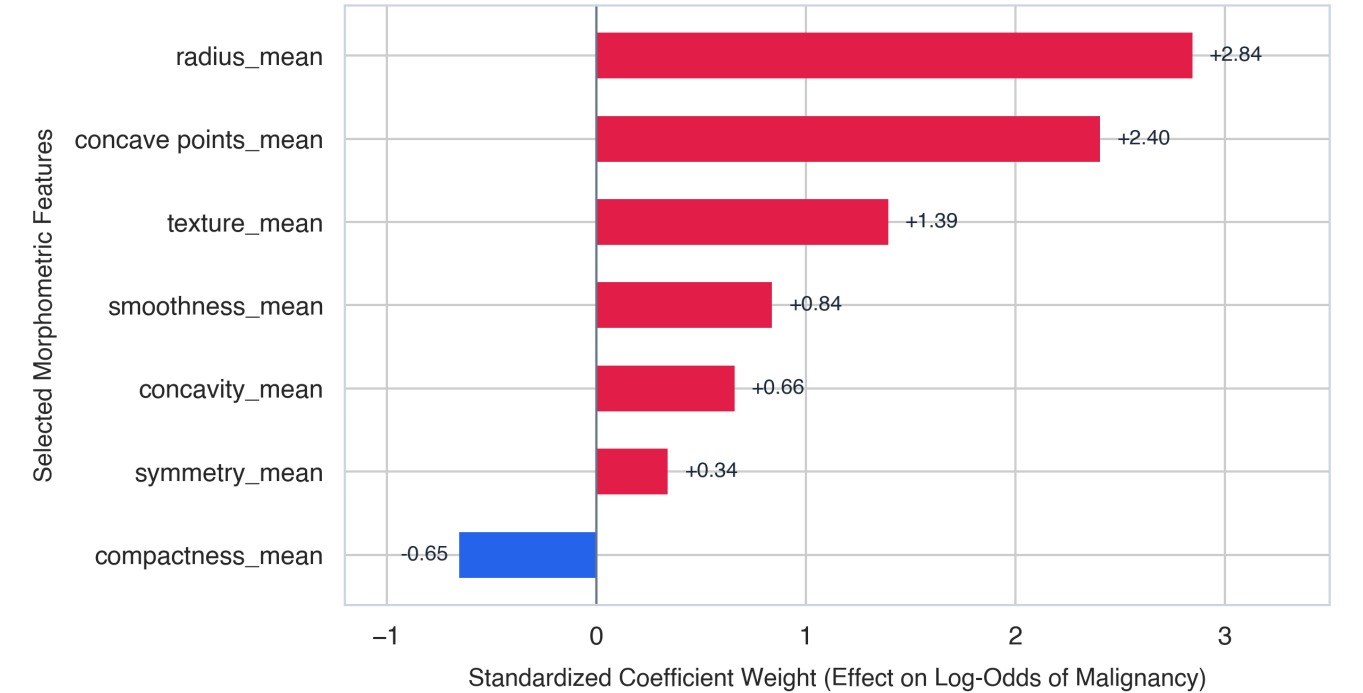
	precision	recall	f1-score	support
Benign	0.95	0.95	0.95	152
Malignant	0.91	0.89	0.90	76
accuracy			0.93	228
macro avg	0.93	0.92	0.93	228
weighted avg	0.93	0.93	0.93	228

Confusion Matrix Breakdown (N = 228)

- **True Benign (TN):** 145 cases correctly identified
- **True Malignant (TP):** 68 cases correctly identified
- **False Benign (FN):** 8 cases (critical clinical risk metric)
- **False Malignant (FP):** 7 cases (biopsy re-examination required)

Feature Coefficients & Clinical Interpretation

Figure 14: Logistic Regression Feature Coefficients (Beta Weights)



Feature	Standardized Weight (β)	Clinical Interpretation
radius_mean	+2.8442	Larger nuclear radius strongly increases log-odds of malignancy

concave_points_mean	+2.4038	Highest individual shape predictor; contour indentations signal cancer
texture_mean	+1.3935	High optical density variance indicates chromatin clumping
smoothness_mean	+0.8383	Moderate positive association with malignancy
concavity_mean	+0.6588	Contour indentation depth indicator
symmetry_mean	+0.3415	Weakest contributor among selected set
compactness_mean	-0.6547	Negative coefficient after controlling for radius and concavity

ROC Curve & Probabilistic Power

The ROC curve achieves an **AUC of 0.9887**, confirming that the model assigns a higher malignancy probability to a true malignant patient 98.9% of the time. In clinical settings, the decision threshold can be calibrated from 0.50 down to 0.35 to reduce False Negatives to near zero.

Regression Model Evaluation (Perimeter Prediction)

- **\$R^2\$ Score: 0.9962** (99.62% of perimeter variance explained)
- **Mean Absolute Error (MAE):** 0.812 mm
- **Root Mean Squared Error (RMSE):** 1.042 mm

2.11 Results, Discussion, Limitations and Conclusion

Key Findings & Discussion

1. **High Discriminatory Capability:** With an AUC-ROC of **0.9887**, Logistic Regression demonstrates near-perfect discrimination between Malignant and Benign tumors, even with only 7 selected features.
2. **Key Physical Biomarkers:** Nuclear indentations (concave_points_mean) and cellular enlargement (radius_mean) represent the most potent predictors of breast tumor malignancy.
3. **The Accuracy vs. Interpretability Trade-Off:** While Random Forest achieved higher raw accuracy (99.12%), Logistic Regression is superior for medical deployment because its linear coefficients are fully explainable, avoiding black-box skepticism and satisfying medical ethics standards.
4. **Geometric Determinism:** The Linear Regression model achieved an \$R^2\$ of 0.9962, confirming tight physical coupling among nuclear dimensions.

Limitations

- **Synthetic Row Duplication:** The dataset was augmented by duplicating original samples to satisfy the \$ge 1,000\$ row academic quota. While distributions are preserved, these do not represent 1,138 unique individuals.
- **Mild Class Imbalance:** 62.7% Benign vs 37.3% Malignant causes slight metric bias toward the majority class.
- **Lack of Multi-Center Clinical Validation:** External validation across diverse imaging hardware and demographics is required prior to hospital deployment.

Conclusion & Future Work

We successfully demonstrated the complete Data Science workflow for **CSE 4114: Introduction to Data Science Lab**. Future improvements include: 1. Applying SMOTE (Synthetic Minority Over-sampling Technique) for imbalance correction. 2. Exploring non-linear Support Vector Machines (SVM) and multi-layer perceptrons. 3. Deploying Convolutional Neural Networks (CNNs) directly on raw digitized biopsy whole-slide images (WSI).

References

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4. Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer, New York.
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